Port-Hamiltonian Control for Grasping with a Unitree Humanoid

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***Abstract*— Humanoid grasping involves contact with objects of unknown geometry, where bounded interaction energy matters as much as tracking accuracy. Port-Hamiltonian (PH) modelling makes a system's energy storage, dissipation, and power exchange explicit, so that, for positive gains and without integral action, a passivity identity follows from the closed-loop structure. This paper presents an energy-based control layer for the twelve proximal finger and thumb joints (HAND12) of the Unitree G1 humanoid, built directly from the robot's own URDF-derived multibody model. The HAND12 joints are governed by a mode-switched interconnection and damping assignment passivity-based control (IDA-PBC) law: an impedance-shaped position controller in free space, and a force-regulating law once contact is established. Both modes are implemented and exercised in simulation, where the settled finger angles depend on the object (two widths tested), and on the physical robot in five grasp-pipeline runs with one object and one arm bench comparison of the PH and position-PD laws. On hardware, the hand accepts only position commands, which a virtual-admittance stage bridges without a passivity guarantee, and one thumb-yaw force sensor is faulty, so that joint is excluded.**

***Index Terms*— Port-Hamiltonian systems, humanoid robotics, impedance control, dexterous manipulation, energy-based control, IDA-PBC, Unitree G1.**

I. Introduction

Humanoid robots must interact physically with their environment while maintaining accurate and safe motion. Classical PID control tracks well, but exposes no energy-based safety certificate. Port-Hamiltonian (PH) systems provide a modelling and control framework in which interconnection, energy storage, dissipation, and power exchange are represented explicitly [1], [2], so that passivity is a property of the controller's structure. This paper applies an implementation-oriented PH control layer to the Unitree G1 humanoid, integrated in Isaac Lab [3] and then carried onto physical hardware.

The system is organised in three parts. First, Section II builds a Port-Hamiltonian model of the G1's twelve proximal finger and thumb joints (HAND12) directly from the robot's own URDF, together with an IDA-PBC law that switches between an impedance-shaped position mode and a force-regulating mode; the arm and waist remain under the existing inverse-kinematics and position-PID stack. Second, Section III adapts the same energy-based structure for the physical robot, where the arm accepts torque commands directly but the hand does not, motivating a virtual-admittance stage that renders the same control law onto the hand's position-servo interface. Third, Sections IV and V evaluate the simulated and the physical controllers in turn: in simulation, through structural, implementation, and grasp experiments; on hardware, through an arm bench comparison and five repeated grasp-pipeline runs.

This model-based, structure-first design contrasts with recent learning-based dexterous-grasping policies [4] and reinforcement-learning-guided whole-body model-predictive control explored on the same G1 platform [5], where safety depends on training data rather than on system structure.

The main contribution is this implementation-oriented energy-based control layer, together with an explicit statement of the assumptions its theoretical guarantees require, and empirical checks of the implemented control law. The formal result (Proposition 1, Section II-D.2) is a passivity identity for the HAND12 subsystem without integral action; convergence is established only for non-controlled joints held fixed, and it is not a whole-robot stability claim.

II. Port-Hamiltonian Modelling and Control: Simulation

A. General Form

An input–state–output PH system is described by

$$\dot{x} = [J(x) - R(x)]\nabla H(x) + g(x)u, \qquad y = g(x)^\top \nabla H(x), \quad (1)$$

where $J(x) = -J(x)^\top$ is the interconnection matrix, $R(x) = R(x)^\top \succeq 0$ is the dissipation matrix, $H(x)$ is the Hamiltonian, and $g(x)$ maps the input to the system. The energy balance is

$$\dot{H} = u^\top y - \nabla H(x)^\top R(x) \nabla H(x) \leq u^\top y. \quad (2)$$

If H is bounded from below, the system is passive with supply rate $u^\top y$ [1].

B. Mechanical Form

For a mechanical system with coordinates $q \in \mathbb{R}^n$ and momenta $p = M(q)\dot{q}$, the Hamiltonian is

$$H(q,p) = \frac{1}{2} p^\top M^{-1}(q) p + U(q), \quad (3)$$

where $M(q) = M(q)^\top > 0$ is the inertia matrix and $U(q)$ is the potential energy. The canonical structure used here is

$$\begin{bmatrix} \dot{q} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} 0 & I \\ -I & -D(q,p) \end{bmatrix} \begin{bmatrix} \partial H/\partial q \\ \partial H/\partial p \end{bmatrix} + \begin{bmatrix} 0 \\ B(q) \end{bmatrix} u. \quad (4)$$

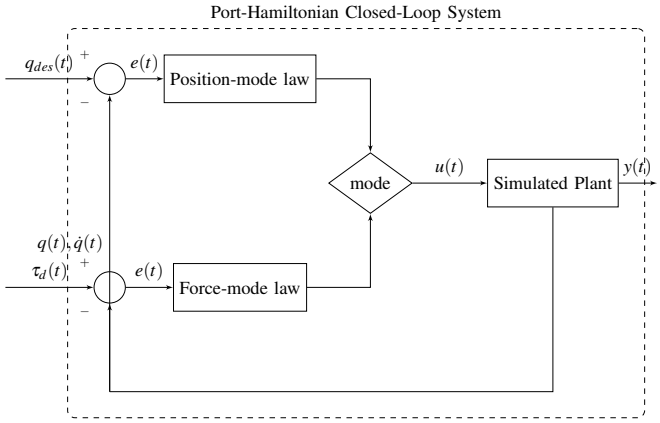


Fig. 1. Two switchable laws with different reference types (a position target $q_{des}(t)$ versus a torque reference $\tau_d(t)$), each closed through its own feedback summation junction, one plant output, all inside one PH boundary.

C. Robot Modelling and Kinematics

A Unified Robot Description Format (URDF) file for the G1 humanoid, provided by Unitree Robotics, is parsed into three data classes: *LinkInertial* (mass, center-of-mass, inertia tensor per link), *JointSpec* (joint type, axes, limits, parent/child relationships), and *URDFModel* (the full connectivity graph). The verified model contains 63 links, 62 joints, and 53 movable joints. The root link is `pelvis`; the fixed-base tree therefore has $N_B = 53$ configurational degrees of freedom, of which twelve are the HAND12 actuator selection $B \in \mathbb{R}^{53\times 12}$.

Featherstone spatial algebra [6], [7] constructs the multibody tree, consistent with recent port-Hamiltonian multibody formulations [8]. The composite-rigid-body algorithm (CRBA) computes the mass matrix $M(q) \in \mathbb{R}^{53\times 53}$ [6], [9]. Section IV-A reports the resulting symmetry, positive-definiteness, and URDF mass-consistency checks.

The remaining seventeen arm-and-waist joints are governed by a separate position-PID law, adapted for hardware in Section III. The three waist joints are part of the dynamic model, entering $M(q)$ and, on hardware, the arm gravity term through the measured waist angle, but they are not a control input of the arm or hand law. In the hardware grasp runs the waist is not a decision variable of the inverse kinematics: it is held at a fixed assumed angle of zero and maintained on hardware by a constant-reference position loop. This was a deliberate choice, since a free waist was recruited by the solver (up to 21.7° of yaw in captured poses) without improving reach; a weighted option ($w = 30$) remains available but was not active in the logged runs.

The hand kinematics use the URDF joint transforms directly. Mimic relationships, characteristic of tendon-driven Inspire-hand designs [10], are represented explicitly: the thumb intermediate and distal joints move at $1.6\times$ and $2.4\times$ the pitch coordinate, and the non-thumb intermediate joints move 1:1 with their proximal coordinate.

For a fingertip position p_{tip}, the geometric Jacobian satisfies $\dot{x}_{\text{tip}} = J(q)\dot{q}$. For a revolute joint with axis $\hat{a}_j$, origin o_j, and mimic scale s_j, its Jacobian columns are $J_v[:,j] = s_j\hat{a}_j \times (p_{\text{tip}} - o_j)$ and $J_\omega[:,j] = s_j\hat{a}_j$. For kinematic cross-validation, a damped least-squares inverse-kinematics (DLS-IK) update is applied to the arm-side chain [11]:

$$\Delta q = J^\top (JJ^\top + \lambda^2 I)^{-1} e, \qquad q_{k+1} = \text{clip}(q_k + \alpha \Delta q). \quad (5)$$

For the twelve HAND12 joints, a closed-form analytical inverse kinematics provides a deterministic alternative to iterative DLS, used to generate contact-consistent grasp targets during segment planning.

D. HAND12 IDA-PBC

HAND12 denotes the twelve proximal finger and thumb joints of the Unitree G1's two hands: the thumb yaw and thumb pitch joints, and the index, middle, ring, and pinky proximal joints, six per hand. These are the joints the Port-Hamiltonian layer of this paper controls; the remaining body joints are governed separately (Section II-C).

The Port-Hamiltonian system follows interconnection and damping assignment (IDA)-based passivity control [1], [12], [13], informed by prior implementations on robotic manipulators and fingers [7], [14]–[17] and by energy-aware impedance designs on other physically interacting platforms [18].

1) Plant and Assumptions: The tree has 53 movable joints: 12 leg, 3 waist, 14 arm (seven per arm), and 24 hand joints. Of the hand joints, twelve are the HAND12 proximal joints q_c driven by the controller, and twelve are unactuated mimic (intermediate and distal) joints. The 15 leg and waist joints are assumed held at a constant configuration: their velocities vanish, their constraint forces do no work, and they drop out of the dynamics of the remaining coordinates. The remaining coordinates are $q = (q_c, q_{nc})$, where the non-controlled joints q_{nc} are the 14 arm joints and the 12 mimic joints. With Hamiltonian $H(q,p) = \frac{1}{2}p^\top M^{-1}(q)p + U(q)$, where M is the corresponding principal submatrix of the 53×53 mass matrix, the plant is

$$\dot{q} = M^{-1}p, \qquad \dot{p} = -\nabla_q H - D\dot{q} + Bu_c + B_{nc}u_{nc} + J_{\text{tip}}^\top F_{\text{contact}}, \quad (6)$$

with u_c the HAND12 torques, u_{nc} the non-controlled torques (arm torques from the separate position-PID law, zero on the mimic joints), $J_{\text{tip}} = \partial x/\partial q$ the fingertip Jacobian, and F_{contact} the force exerted on the fingertips by the environment. The structure matrices are

$$J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \qquad R = \text{blkdiag}(0, D), \qquad G = \begin{bmatrix} 0 \\ B \end{bmatrix}. \quad (7)$$

The natural HAND12 output is $y = G^\top \nabla H = B^\top \dot{q} = \dot{q}_c$.

The analysis assumes $M(q) > 0$, a differentiable gravitational potential, diagonal gains $K_p, K_d \succ 0$, non-negative physical damping, and, for Proposition 1, no integral action ($K_i = 0$); the integral term used in simulation lies outside the proposition. The result concerns the HAND12 subsystem: the arm and mimic joints are non-controlled and enter only through the port stated below, so it does not establish stability of the fully moving robot. Joint positions are in rad and torques in Nm throughout, so K_p and K_{spr} (Section II-D.3) have units Nm/rad, K_d has units Nm·s/rad, and K_i has units Nm/(rad·s).

The twelve HAND12 joints are governed by one of two mutually exclusive control laws, selected by a segment-driven state machine: a position-mode law (Section II-D.2) during free-space reaching and pregrasp segments, and a force-mode law (Section II-D.3) once contact is detected or scheduled.

2) Position Mode: Let $e = q_c - q_c^\star$, with a constant target $q_c^\star$ (piecewise constant between segment switches), and let z be the integral state with $\dot{z} = e$. The implemented position-mode control law is

$$u_c = g_c(q) - K_p e - K_d \dot{q}_c - K_i z, \qquad g_c(q) = \frac{\partial U}{\partial q_c}. \tag{8}$$

In simulation the default gains are $K_p = 4\,\text{Nm/rad}$ ($1.0\,\text{Nm/rad}$ on thumb-yaw), $K_d = 1\,\text{Nm·s/rad}$, and $K_i = 0.1\,\text{Nm/(rad·s)}$, with the integral state clamped to $|z| \le 0.35\,\text{rad·s}$. With exact gravity matching, the gravity torque on the HAND12 joints is cancelled and the remaining torque is the impedance term $-K_p e - K_d \dot{q}_c - K_i z$. For the proposition, take $K_i = 0$ and the energy

$$W = \tfrac{1}{2} p^\top M^{-1}(q) p + \tfrac{1}{2} e^\top K_p e, \tag{9}$$

the kinetic energy plus the virtual spring; the potential U is deliberately not included, because it is cancelled by g_c on the HAND12 joints.

Proposition 1. Under the stated assumptions and in free space ($F_{\text{contact}} = 0$, as in position mode), along trajectories of the position-mode closed loop,

$$\dot{W} = -\dot{q}^\top D \dot{q} - \dot{q}_c^\top K_d \dot{q}_c + \dot{q}_{nc}^\top \big(u_{nc} - \nabla_{q_{nc}} U(q)\big). \tag{10}$$

The closed loop is therefore passive from the net non-controlled torque $u_{nc} - \nabla_{q_{nc}} U$ to $\dot{q}_{nc}$, with storage $W \ge 0$.

Proof. With $\dot{M} - 2C$ skew-symmetric, the kinetic energy satisfies $\dot{K} = \dot{q}^\top \big(B u_c + B_{nc} u_{nc} - \nabla_q U - D\dot{q}\big)$. The HAND12 part of $\nabla_q U$ equals g_c, so substituting (8) with $K_i = 0$ gives $\dot{q}_c^\top (u_c - g_c) = -\dot{q}_c^\top K_p e - \dot{q}_c^\top K_d \dot{q}_c$. The first term cancels $\frac{d}{dt}\frac{1}{2} e^\top K_p e = e^\top K_p \dot{q}_c$ because K_p is symmetric, leaving (10). ■

Since $D \succeq 0$ and $K_d \succ 0$, (10) gives $\dot{W} \le \dot{q}_{nc}^\top (u_{nc} - \nabla_{q_{nc}} U)$. If the non-controlled joints are held fixed ($\dot{q}_{nc} \equiv 0$), W is positive definite in (p_c, e) and $\dot{W} \le 0$, so the desired HAND12 configuration is stable in the sense of Lyapunov; the sublevel sets of W are compact because $K_p \succ 0$ and e is bounded by the joint limits, and LaSalle's invariance principle gives $\dot{q}_c \to 0$ and then $e \to 0$. This requires a constant target, $K_d \succ 0$, no contact, and no torque saturation (Section VI). The idealization fixes the arm joints and also the mimic-coupled intermediate and distal finger joints, which are non-controlled degrees of freedom of the tree that move with their proximal joints; a statement for the coupled finger needs mimic-eliminated coordinates and the corresponding reduced gravity gradient.

The non-controlled joints are not fixed on the physical robot, so the convergence statement is an idealization; Section VI discusses the waist and the arms.

3) Force Mode: Force-mode control [19] shapes a virtual spring torque $\tau_{\text{spring}} = K_{\text{spr}} \odot (q_c - q_{0,\text{spr}})$ against a desired torque τ_d, with force-shaping potential

$$V_d = \frac{1}{2} \sum_i \frac{K_{f,i}}{K_{\text{spr},i}} (\tau_{\text{spring},i} - \tau_{d,i})^2, \; \nabla_{q_c} V_d = K_f \odot (\tau_{\text{spring}} - \tau_d). \tag{11}$$

With an optional task-space potential and damping matrix, the force-mode law is

$$u_{\text{force}} = g_c(q) - K_f \odot (\tau_{\text{spring}} - \tau_d) - K_d \dot{q}_c \tag{12}$$

$$+ J_c^\top [K_t (x_d - x) - B_t J_c \dot{q}_c]. \tag{13}$$

By default $K_{\text{spr}} = 50\,\text{Nm/rad}$; K_f is dimensionless, since $(\tau_{\text{spring}} - \tau_d)$ already carries the torque units. The task-space gains default to $K_t = 5\,\text{N/m}$, $B_t = 0.5\,\text{N·s/m}$ in free space, and $K_{t,\text{grasp}} = 250\,\text{N/m}$, $B_{t,\text{grasp}} = 12\,\text{N·s/m}$ during grasp, lift, and transport.

Within intervals where the task-space gains and the targets x_d and τ_d are constant, take the energy $W_f = \frac{1}{2} p^\top M^{-1}(q) p + V_d + \frac{1}{2}(x_d - x)^\top K_t (x_d - x)$, which, as in (9), excludes U because g_c cancels it on the HAND12 joints. The computation of Proposition 1 then gives

$$\begin{aligned} \dot{W}_f = & -\dot{q}^\top D \dot{q} - \dot{q}_c^\top K_d \dot{q}_c - \dot{q}_c^\top J_c^\top B_t J_c \dot{q}_c + e_I^\top f_I \\ & + \dot{q}_{nc}^\top \big(u_{nc} - \nabla_{q_{nc}} U - J_{nc}^\top K_t (x_d - x) + J_{nc}^\top F_{\text{contact}}\big), \end{aligned} \tag{14}$$

with $e_I = F_{\text{contact}}$ the force exerted on the fingertips by the environment, $f_I = J_c \dot{q}_c$, and $J_{nc} = \partial x / \partial q_{nc}$, since x is expressed in the pelvis frame. This establishes passivity at the contact port between switching instants. The non-controlled port also carries the task spring and the contact force transmitted through the non-controlled joints; it reduces to that of Proposition 1 when $\dot{q}_{nc} = 0$. Here J_c is the fingertip Jacobian with respect to the HAND12 joints with the mimic coupling propagated, so (14) holds under that constraint and only approximately in simulation, where the mimic joints are separate degrees of freedom; the task-space term is inactive on hardware. The task-space gains (K_t, B_t) are themselves switched by a live contact-residual test and by the active state-machine segment, so (14) does not by itself cover the switching instants; Section IV-A reports a direct numerical check of the implemented law that is not subject to this restriction.

E. Isaac Lab Integration

The Isaac Lab [20] grasping environment builds on a G1 simulation foundation originally developed by Vinke [3], extended here with the PH finger layer above. The PH controller runs behind a runtime flag, leaving the existing seventeen-joint body controller unaffected. At each control step (60 Hz), the controller maps simulator joint states to the PH tree ordering by name, computes gravity compensation and damping injection, and writes the twelve HAND12 torques into the full joint-effort vector. Isaac Lab is used here only to execute and validate this analytically derived controller, unlike sampling-based control that uses a parallelizable simulator directly as the dynamics model [21].

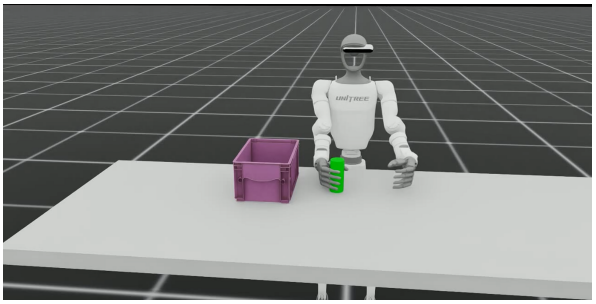


Fig. 2. The G1 humanoid and cube grasp scene as rendered inside Isaac Lab, used to test and validate the HAND12 controller.

III. Port-Hamiltonian Modelling and Control: Physical Hardware

The gap between simulation and hardware is not uniform across joints, and it determines how the laws of Section II are rendered onto the physical robot.

Arm. The Unitree `rt/arm_sdk` interface accepts torque commands directly, so a law with the same gravity-compensation-plus-impedance structure as the position-mode law (8), without the integral term, is sent to the physical arm largely unchanged. The per-joint gains used on hardware are $K_{p,\mathrm{ph}} = [15,15,10,10,4,3,3]$ Nm/rad and $K_{d,\mathrm{ph}} = [1.2,1.2,0.8,0.8,0.4,0.3,0.3]$ Nm·s/rad (shoulder pitch/roll/yaw, elbow, wrist roll/pitch/yaw), with the commanded torque clipped to $0.4\times$ the URDF effort limit per joint. These values, and the position-PID baseline's own firmware gains, are nominally in Nm/rad and Nm·s/rad, but are Unitree `rt/arm_sdk MotorCmd_` fields rather than independently SI-calibrated gains. In PH mode the firmware position loop is lowered to a floor of $k_p = 3$ Nm/rad, $k_d = 0.3$ Nm·s/rad, so the holding torque must come from the commanded torque; Section V-A verifies that it is applied.

Hand. The Inspire hand's Modbus interface, in contrast, exposes only a position servo, with no torque channel at all. The position- or force-mode law ((8), (13)) still computes a torque $\tau_{ideal}(t)$, but with hardware gains that differ from simulation, as configured in the code: $K_p = 12$ Nm/rad (thumb yaw 4), $K_i = 0.1$ Nm/(rad·s), $K_d = 1$ Nm·s/rad, $K_f = 2$, and $K_{\mathrm{spr}} = 1.5$ Nm/rad (thumb pitch 14), against $K_{\mathrm{spr}} = 50$ Nm/rad in simulation. The task-space term is not active on hardware, and the hand loop runs at 5 Hz. This torque cannot be sent to the hand directly. It is instead rendered through a virtual-admittance stage,

$$M\ddot{q}_{cmd} + B\dot{q}_{cmd} + K_{\mathrm{track}}(q_{cmd} - q_{act}) = \tau_{ideal} + \tau_{contact}, \quad (15)$$

applied independently per joint, with the same scalar $M = 7.0\,\mathrm{kg\cdot m^2}$, $B = 12.0$ Nm·s/rad, and $K_{\mathrm{track}} = 2.0$ Nm/rad broadcast to each of the twelve HAND12 degrees of freedom rather than coupling them, before the resulting position command $q_{cmd}(t)$ reaches the hand plant. A contact gain of 5.0 Nm/N converts the force excess above a guard band, set at 0.75 of the 300 g Modbus force limit, into $\tau_{contact}$ (grasp-pipeline defaults, not bench-tuned). Reported forces use the Modbus interface's native gram unit, converted via 9.80665×10^{-3} N/g.

The HAND12 impedance controller therefore cannot actuate the hand directly on hardware, only through this intermediate position-command layer: a sim-real gap specific to the hand, and absent from the arm. Section V reports results under both laws without re-deriving them.

IV. Simulation Validation and Results

The validation pipeline checks the implementation in stages, from static structural properties of the model through to closed-loop behavior under contact.

A. Structural and Implementation Checks

The URDF and index checks verify joint-name alignment rather than relying on assumed simulator ordering. The CRBA and PH-wrapper checks, evaluated on the actual CRBA and forward-kinematics implementation rather than a reduced symbolic model, give 53 movable of 62 joints, a total URDF link mass of 33.38 kg against a tree body mass of 29.57 kg, an exactly symmetric and positive-definite $M(0)$, $\|\nabla_p H - M^{-1}p\| = 9.3 \times 10^{-14}$, and a power-balance relative error of 4.8×10^{-6}.

The force-mode validator checks gain positivity, agreement between the implemented law and its closed form, the spring-potential gradient against finite differences, and fingertip-Jacobian sparsity; all pass. The tightest check evaluates the full task-space balance $u + \nabla_{q_c} V_d + K_d\dot{q}_c - g_c - [J_c^\top K_t(x_d - x) + J_c^\top B_t J_c \dot{q}_c] = 0$ on the running controller, with the task-space term active as configured during lift and transport ($K_{t,\mathrm{grasp}} = 250$ N/m, $B_{t,\mathrm{grasp}} = 12$ N·s/m): the maximum residual over the twelve HAND12 joints is 7.77×10^{-16}, against a requested threshold of 10^{-10}. This confirms that the implemented control law is the one (13) describes; it does not test passivity.

B. Mode-Switching Energy Transient

The per-finger torque rail is set to 10 Nm, matching the Inspire hand's rated capability. In the resulting energy budget no ideal torque exceeds the rail, the applied and designed energy agree (1.69 J each), and the saturator removes none (Fig. 3). Once the grasp, lift, and transport segments complete, HAND12 switches back to the position-mode law (Section II-D.2) with a reference commanding the fingers fully open: the non-thumb joints go to their URDF lower limit of 0 rad, and the thumb yaw and pitch joints to their URDF lower limit of -0.1 rad, identically on both hands.

The largest remaining torque discontinuity occurs at the force-to-position mode transition at release (S6): a 10.75 Nm jump reaching 80.22 W peak power over a four-sample window, carrying more than half the run's total applied energy. This transient results from the controller switching from force regulation back to position tracking while a large position error is still present; the existing transition-blending mechanism smooths the earlier position-to-force transitions but does not yet cover this one. Section VI discusses it as an open item.

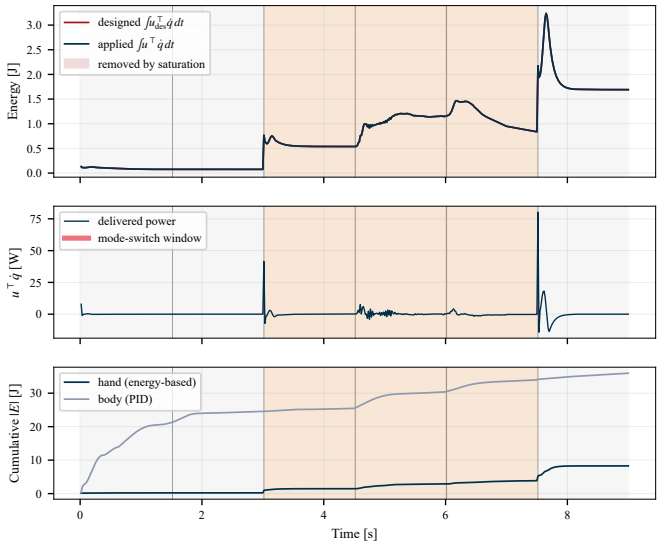


Fig. 3. Energy ledger at the rated 10 Nm torque rail: applied and designed energy track each other exactly.

C. Object-Width Sweep: Impedance-Driven Grasp Equilibrium

To test whether the settled grasp configuration is impedance-driven rather than a fixed scripted target, the desired contact torque is held fixed while the grasped object's width is varied. The settled right-hand joint equilibria (last 50 % of the lift and transport segments) are compared across widths of 25 mm and 40 mm.

TABLE I

RIGHT-HAND GRIP-JOINT EQUILIBRIUM VS. OBJECT WIDTH (25 MM VS. 40 MM).

Joint	Slope [mrad/mm]	Shifted?
Thumb pitch	−27.56	Yes
Index proximal	−6.43	No
Middle proximal	−38.59	Yes
Ring proximal	−35.42	Yes
Pinky proximal	−26.37	Yes

Four of five right-hand grip joints (excluding the orientation-only thumb-yaw joint) shift equilibrium angle by more than three times their own settled-window jitter, with a median slope of 27.6 mrad/mm. At 25 mm these fingers close to their 1.7 rad joint limit; at 40 mm the object stops them short of it. The settled configuration therefore depends on the object rather than on a scripted target, as expected for the shifted joint spring of the force mode. Two widths with one run each were tested.

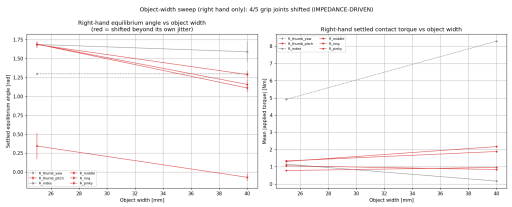


Fig. 4. Settled right-hand grip-joint equilibrium angle (left) and contact torque (right) versus grasped object width, 25–40 mm, corresponding to Table I.

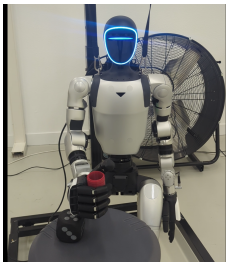


Fig. 5. Physical Unitree G1 lab setup: the right hand holding a rigid cylindrical object during a grasp-hold segment, with the left arm not actuated.

A diagnostic run helps explain why thumb-yaw is excluded above. Over the grasp, lift, and transport segments, thumb-yaw's measured angle sits pinned at its 1.3 rad mechanical limit while its commanded torque grows monotonically, reaching a peak of 7.24 Nm at the exact instant the joint is at that limit, rather than settling as the other four fingers do. In simulation, the force-mode virtual spring fails to locate a stable equilibrium for thumb-yaw within its travel range.

D. Comparison with a Classical Baseline

Before the PH layer of Section II-D, the same Isaac Lab G1 environment [3] drove HAND12 with a classical PD/impedance law, tuned by manual gain sweep rather than derived from the model. Right-hand HAND12 position-tracking error, measured against the commanded position, improves under that empirical tuning pass from an RMSE of 0.23–0.26 rad to 0.008–0.017 rad on the four non-thumb joints (maxima 0.60–0.65 rad to 0.07–0.14 rad), and from 0.074 to 0.044 rad on thumb pitch, while thumb yaw stays at 0.434 rad (maximum 1.620 rad) [3].

The thumb-yaw maximum error (1.620 rad) exceeds the joint's 1.4 rad range, so the reference left the joint limits; gain tuning cannot remove such an error.

The PH controller derives its structure from the plant model (its gains were chosen by hand), its implementation matches the closed form to 7.77×10^{-16} (Section IV-A), and it closes the hand and lifts the object in simulation. The comparison is not like-for-like: the baseline runs saturate at 1.0 Nm, the tasks differ (steady holds against larger open/close motions), and no PH tracking-error table is reported. Thumb-yaw's own force-mode equilibrium shows a further, unresolved limitation on that joint.

V. PHYSICAL VALIDATION

This section reports results from the physical Unitree G1: a right-arm-only bench test isolating the arm's torque law, and five independent runs of the full grasp pipeline, executed with the right arm and hand.

A. Arm Bench-Test Tracking

The same choreography (ramp-in, move-to-pose, hold, return, ramp-out) is run back-to-back, once under the PH arm law of Section III and once under the baseline position-PID; only the hold segment's control law differs between the two runs, so both reach the pose under the same position law and the hold keeps the state that approach left. Logs are recorded

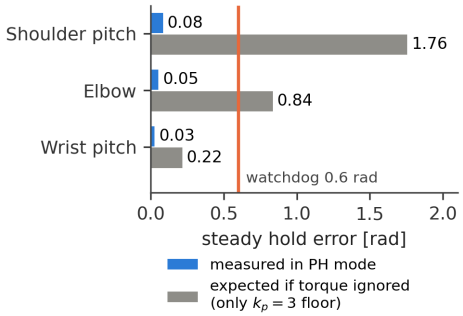


Fig. 6. Steady hold error on the physical arm in PH mode (blue) against the error expected if the firmware ignored the commanded torque and only the $k_p = 3$ Nm/rad floor held the pose (gray). The 0.6 rad watchdog limit is marked.

at 10 Hz while the controller runs at 50 Hz, and hold errors are averaged over the hold segment after discarding its first second; both holds are static, with errors constant to about 10^{-4} rad.

Torque channel. If `rt/arm_sdk` ignored the commanded torque, the $k_p = 3$ Nm/rad floor alone would leave hold errors of 1.76, 0.84, and 0.22 rad on shoulder pitch, elbow, and wrist pitch, the first two beyond the 0.6 rad watchdog limit. The measured errors are 0.084, 0.052, and 0.026 rad and the watchdog never triggered (Fig. 6). The arm, unlike the hand, therefore accepts and applies a torque-level PH law directly. The motors' torque estimate (derived from motor current, so not an independent calibration) agrees with the inferred holding torque within 3–11 %.

PH vs. position-PID. Figure 7 and Table II compare the two holds. The PH hold errors are 0.084 and 0.052 rad on shoulder pitch and elbow, against 0.085 and 0.049 rad for the baseline, and 0.026 rad on wrist pitch against 0.019 rad, with total stiffness $K_{p,\mathrm{ph}} + 3 = 18$, 13, and 6 Nm/rad against 60 Nm/rad. The PH law feeds forward the modeled gravity torque, but the motors' torque estimate during the PH hold is 1.24–1.35× the Pinocchio [22] estimate $g(q)$ (Table II), and this residual 24–35 % is not attributed: friction, remaining model error, and encoder offset are not separated. With this residual, equal static errors at the two stiffnesses are what the arithmetic gives on shoulder pitch and elbow, so the pair of runs, one per law, does not rank the two laws. Neglecting friction, a PH law with matched total stiffness is *predicted* to settle at 0.022, 0.007–0.019, and 0.002–0.011 rad, depending on which run's holding torque is used; no matched-stiffness hardware run has been performed.

Figure 8 shows both responses against the commanded target: the laws follow the same ramps and both settle short of the target in the hold by a similar margin.

B. Grasp-Pipeline Repeatability

Five independent runs of the full grasp-pipeline state machine (approach through retreat) are executed under the Port-Hamiltonian controller of Section II-D, with the HAND12 joints rendered through the virtual-admittance stage (15) of Section III, and the arm held by the firmware position loop. The commanded force-mode reference is $\tau_d = 0.75$ Nm on

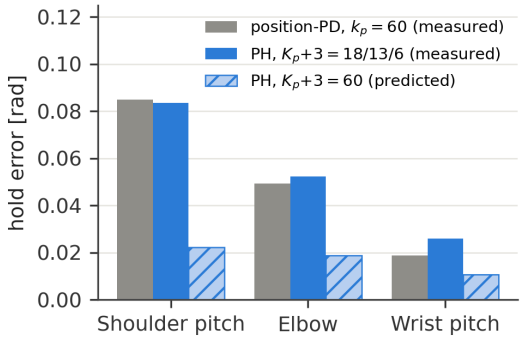


Fig. 7. Hold error of the physical arm under the position-PID baseline (gray) and the PH law (blue), both measured, and the *predicted* PH error at matched total stiffness (hatched, from the baseline run's holding torque).

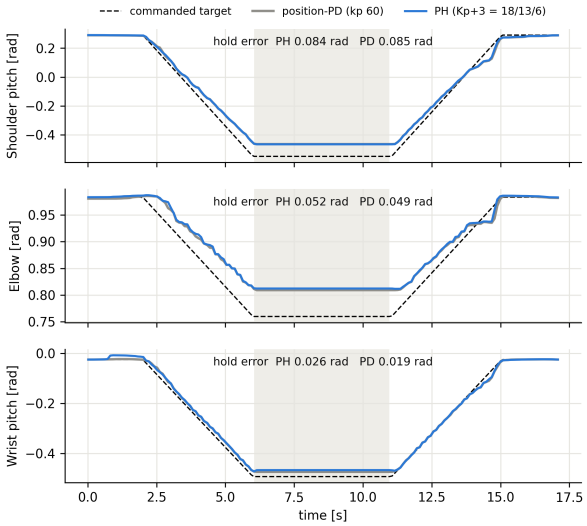


Fig. 8. Right-arm joint response in the arm bench test under the position-PID baseline (gray) and the PH law (blue), against the commanded target (dashed). The shaded band marks the hold segment, where the control law differs; the hold errors are annotated.

each of the six right-hand joints (0.5 Nm base reference scaled by 1.5 for grasp); thumb-yaw is not analyzed here. All runs use one rigid object at one width.

Table III reports the settled grasp-hold state, mean ± standard deviation across the five runs, and Fig. 10 shows the finger angles and contact forces over one representative run. The thumb-yaw joint is excluded: its force sensor is faulty on this unit, independent of the control law, with reported forces spiking to 5923 g over the run, well past the 1000 g fault threshold (9.81 N) used in the pipeline (Fig. 10, bottom).

TABLE III

SETTLED GRASP-HOLD STATE ACROSS 5 RUNS (MEAN ± STD).

Joint	Position [rad]	Force [N]
Thumb pitch	0.113 ± 0.001	0.40 ± 0.05
Index	1.343 ± 0.122	2.09 ± 0.26
Middle	1.408 ± 0.107	2.20 ± 0.86
Ring	1.399 ± 0.106	2.51 ± 1.38
Pinky	0.900 ± 0.699	2.49 ± 0.62

Thumb pitch and index settle tightly across runs (relative

TABLE II

HOLD-SEGMENT ERROR AND PH-MODE TORQUE. $g(q)$ COMES FROM A SEPARATE PINOCCHIO-BASED PIPELINE, NOT THE CRBA CONSTRUCTION OF SECTION II-C; τ_{est} IS THE MOTORS' TORQUE ESTIMATE.

Joint	PH [rad]	PID [rad]	$g(q)$ [Nm]	τ_{est} [Nm]	τ_{est}/g
Shoulder pitch	0.084	0.085	3.76	5.06	1.35
Elbow	0.052	0.049	1.82	2.26	1.24
Wrist pitch	0.026	0.019	0.49	0.61	1.24



Fig. 9. The same force-mode law, computing the same $\tau_{ideal}(t)$ passed to the virtual admittance stage (15), closes on an irregularly shaped foam object without object-specific retuning. The hand is held by an operator off the robot, so this is a qualitative demonstration.

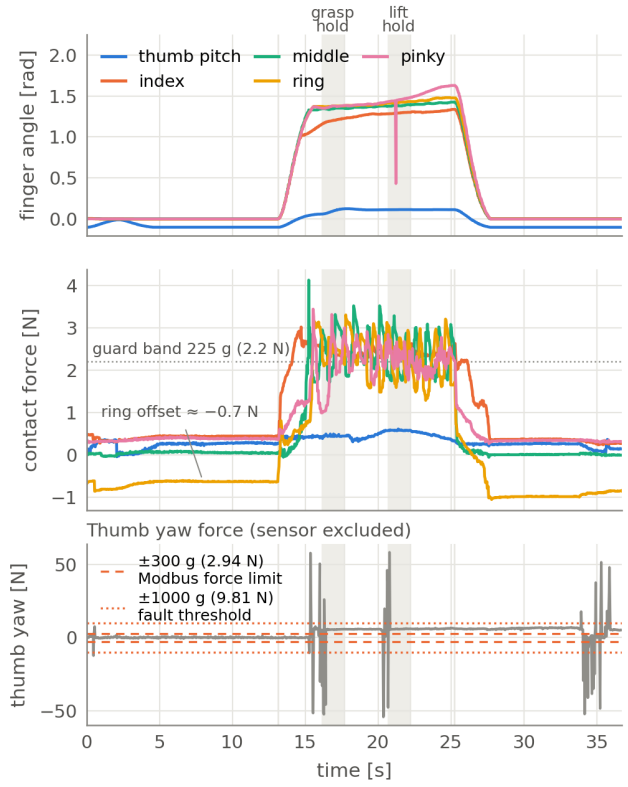


Fig. 10. Representative right-hand grasp run: finger angle (top) and measured contact force (middle) for the five healthy grip joints, and thumb-yaw reported force (bottom, sensor excluded). Shaded bands mark grasp-hold and lift-hold. Dashed: 300 g Modbus force limit; dotted: 1000 g fault threshold; the 225 g guard band is marked in the middle panel.

standard deviation of contact force ≈12–13 %); middle, ring, and pinky show substantially more run-to-run spread (25–55 %). The pinky stayed open in two runs (0.051 and 0.226 rad) against ≈ 1.38 rad in the other three, yet reported about 3 N while open. The settled forces (2.1–2.5 N) cluster around the 225 g (2.2 N) guard threshold of the admittance stage, so on hardware the guard, not τ_d, sets the settled force level, and peak forces exceeded 300 g in every run (maximum 941 g, ring). With one object at one width, these runs cannot test the width dependence seen in simulation.

The final `side_down` segment is excluded from the tracking analysis: the `rt/arm_sdk` blend weight ramps to zero and the arm is handed back to Unitree's on-board controller.

VI. LIMITATIONS

The fixed-base model omits the floating-base six-degree-of-freedom root. Gravity gradients are computed with finite differences, so exact cancellation is replaced by an approximation of order $\mathcal{O}(\varepsilon^2)$. The intermediate and distal hand joints are not algebraically eliminated from the full multi-body tree, although their mimic relationships are included in the kinematics layer.

The formal result requires strictly positive damping gains, whereas runtime gain checks permit zero values. Both the simulated and the hardware hand laws use integral action ($K_i = 0.1$ Nm/(rad·s), clamped), which Proposition 1 excludes, so the theoretical argument does not cover the integral term. The task-space gains change with contact state and state-machine segment, so the force-mode passivity equation (14) is established only between switching instants; no common Lyapunov function covers the hybrid position/force transitions as a whole. Section IV-B's unresolved release transient (S6) is a concrete instance of this gap. Raising the torque rail to match hardware capability reduces how often the actuator saturates but does not establish what happens to the passivity argument when it still does; a sector-nonlinearity or input-to-state-stability treatment of the remaining saturation is future work.

The convergence statement of Proposition 1 assumes that the leg and waist joints, and for convergence all non-controlled joints, are held fixed. In simulation the arms move during the grasp, and on the physical robot the waist is not fixed: it is held at a constant reference by a position loop ($k_p = 60$ Nm/rad) without gravity compensation and leans in pitch by up to 0.043 rad during the arm bench test, identically under the PH and position-PD arm laws. The inverse kinematics assumes a zero waist angle, so this lean displaces the right wrist by about 13 mm unobserved. The passivity identity (10) retains the non-controlled port and remains valid, but no stability claim is made for the moving waist or the position-held left arm. A Lyapunov treatment would need a positive effective stiffness, $K_{nc} + \nabla^2_{q_{nc}} U \succ 0$, and a bound on the load coupling from the arm pose, which is not derived here. On the URDF model the waist satisfies the stiffness condition (about 39 Nm/rad against a gravity

curvature of about $-21\,\mathrm{Nm/rad}$), but the model predicts $0.12\,\mathrm{rad}$ of lean where $0.036\,\mathrm{rad}$ is measured; the cause is not separated, and legs and balance are not modeled.

Finally, the Inspire hand hardware interface exposes a position servo rather than the direct torque interface the PH analysis assumes. Section III bridges this gap with a virtual-admittance stage rendering the same position- or force-mode law onto the hand's position-servo interface; the admittance gains are not bench-tuned, and the passivity argument of Section II-D is stated for the direct-torque interface only. Whether it extends through the admittance stage to the position-servo-driven closed loop has not been established.

VII. Conclusion

This paper presents an implementation of the Port-Hamiltonian framework, developing a force-impedance control strategy for grasping. A passivity identity is derived for the HAND12 subsystem under stated assumptions, the implemented law is checked numerically against its closed form in simulation, and the controller is then evaluated on the physical robot.

In simulation, the settled finger angles depend on the grasped object rather than on a scripted target (two widths, one run each). The PH controller derives its structure from the plant model, with hand-chosen gains; a classical PD/impedance baseline from prior work is reported without a like-for-like PH comparison, and thumb-yaw fails to reach a stable force equilibrium under both approaches, for different reasons.

On hardware, five grasp-pipeline runs with one rigid object at one width are consistent with an impedance-driven equilibrium, and a qualitative demonstration shows the hand closing on an irregularly shaped foam object without object-specific retuning. Two sim-to-real gaps are identified: the Inspire hand's position-only interface requires the virtual-admittance stage of Section III rather than direct torque, and one hand joint's contact sensing is unreliable on this physical unit. In the arm bench test, the motors' torque estimate follows the commanded PH torque, so `rt/arm_sdk` applies a torque-level law. With one run per law, both holds remained static at the pose left by the shared position-PD approach, within $0.003\,\mathrm{rad}$ on shoulder pitch and elbow and $0.007\,\mathrm{rad}$ larger for PH on wrist pitch; the comparison does not rank the two laws, and a residual 24–35 % gap between the motors' torque estimate and the gravity model remains unexplained.

Future work includes extending the passivity argument through the admittance stage, covering the force-to-position mode transition within the hybrid stability argument, and a matched-stiffness hardware comparison of the PH and position-PID arm laws.

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